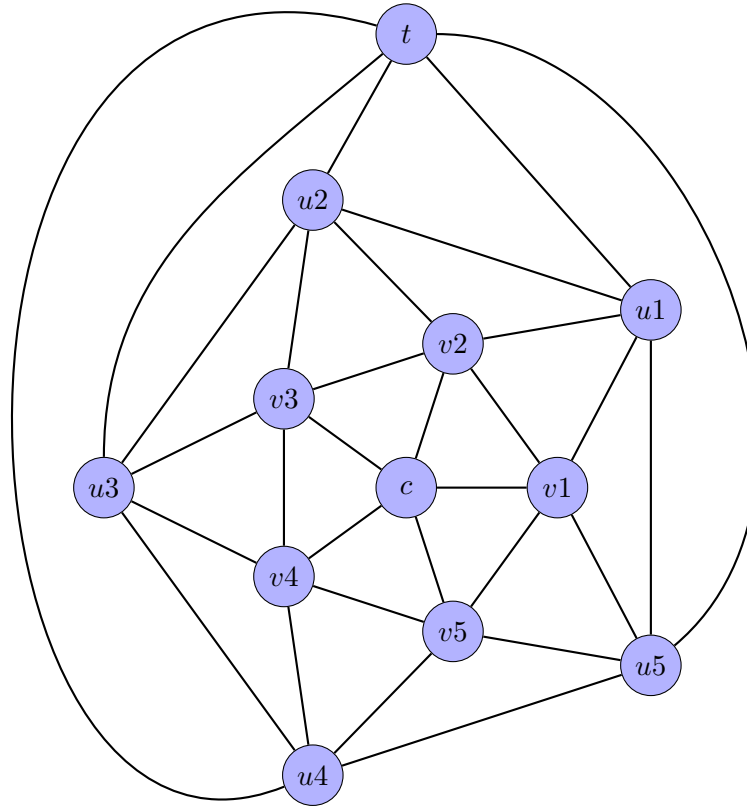


Question 1: Give an example of a planar graph in which no vertex has degree less than 5



Question 2: If G is a planar graph of order at least 4, prove that G contains at least four vertices of degree at most 5

We may assume that G is maximal planar, since if there are at least 4 vertices of degree at most 5 in H , a maximal planar extension of G , then the same applies for G . Let N be the set of vertices of degree at most 5. Then,

$$\sum_{v \in V} \deg v = \sum_{v \in N} \deg v + \sum_{v \in V \setminus N} \deg v \geq |N| \cdot 3 + (n - |N|) \cdot 6 = 6n - 3|N|$$

By the Handshaking Lemma,

$$\sum_{v \in V} \deg v = 2m = 6n - 12$$

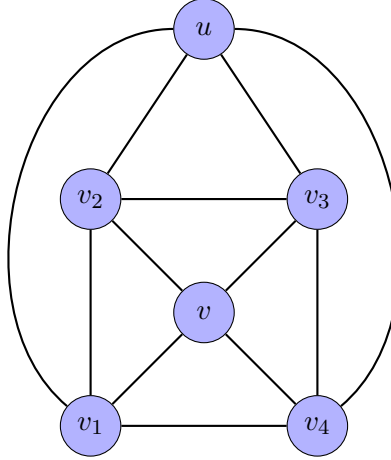
So

$$6n - 12 \geq 6n - 3|N| \iff |N| \geq 4$$

as required.

Question 3: Prove that there exists only one 4-regular maximal planar graph

Suppose G is a 4-regular maximal planar graph with order n . Then every vertex has degree 4, so, by the Handshaking Lemma, $2m = 4n \iff m = 2n$. But since G is maximal planar, we know $m = 3n - 6$. So we have that $3n - 6 = 2n \iff n = 6$. Take any vertex $v \in V(G)$ and let $N(u) = \{v_1, v_2, v_3, v_4\}$. Then, without loss of generality, there is a cycle $C : v_1, v_2, v_3, v_4, v_1$. Then there is one vertex $u \in V(G)$ such that $u \neq v$ and $u \notin N(v)$. Then, since $\deg u = 4$, and u is not a neighbour to v , we must have that $uv_i \in E(G)$ for all $i \in \{1, 2, 3, 4\}$ (all of u 's neighbours are the neighbours of v). This shows that G must be unique, up to isomorphism, and is a graph looking as such:



Question 4: Let $k \geq 3$ be an integer and let G be a plane graph of order $n \geq k$ and size m in which each cycle has length at least k .

(a) Determine an upper bound B for m in terms of n and k .

Clearly $m \leq 3n - 6$. We use this bound if G has no cycle (as otherwise k can be any integer and our equation means nothing). If G has a cycle, let $\epsilon(F)$ be the number of edges in $G[F]$. Then

$$\sum_F \epsilon(F) \leq 2m$$

(we might have bridges that only count an edge once). But for each face F , the number of edges in $G[F]$ is greater than k (as these edges form a cycle). So $2m \geq fk$. By Euler's Formula,

$$2m \geq fk \iff 2m \geq (2 + m - n)k \iff 2m \geq 2k + mk - nk \iff m \leq \frac{k}{k-2}(n-2)$$

Since m is an integer, we can further reduce this to

$$m \leq \left\lfloor \frac{k}{k-2}(n-2) \right\rfloor$$

(this upper bound is better because $k \geq 3$)

(b) Show that the bound B obtained in the previous question is sharp by determining, for arbitrary $k \geq 3$, a plane graph G of order n and size $m = B$ in which every cycle has length at least k .

Consider graph C_k . Every cycle has length k (as there is only one cycle), $n = m = k$. So our upper bound is

$$B = \left\lfloor \frac{k}{k-2}(k-2) \right\rfloor = k = m$$

as required.

Question 5: If G is a plane graph and F is any face of G , prove that G can be embedded in the plane in such a way that F is the outer face.

We prove this by induction on the number of faces. If $f = 1$, then the result follows trivially as it is the external face. So assume that, for $f \geq 1$, this result holds and let G be a graph with $f + 1$ faces. Then $f + 1 \geq 2$ and G is not a tree. Let C be the cycle in the boundary of the external face f_1 . Let e be an edge on C , and f_2 the face that combines with f_1 in $G - e$. $G - e$ has $f - 1$ faces, so every face can be embedded as the outer face. Thus, every face in G except f_2 can be embedded as the outer face. We thus need to show f_2 can be embedded as the outer face. Let C' be the cycle bounding f_2 , then write the cycle on the external of graph $G - e$ (so C' contains $G - e$). This is then an embedding with f_2 as the external face, as required.

Question 6: Euler's Formula applies only to connected planar graphs. Generalize Euler's Formula so that it can be used with any planar graph

Suppose G is a planar graph with components $C_1, C_2, \dots, C_N$. Each component C_i , as a subgraph, has $n_i - m_i + f_i = 2$ for $i \in \{1, 2, \dots, N\}$. But then in G , we have

$$\begin{aligned} n_G &= \sum_{i=1}^N n_i \\ m_G &= \sum_{i=1}^N m_i \\ f_G &= \sum_{i=1}^N f_i - (N - 1) \end{aligned}$$

Since in the faces, we count the outer face N times (one for each component). So

$$n_G - m_G + f_G = \sum_{i=1}^N (n_i - m_i + f_i) - (N - 1) = 2N - N + 1 = N + 1$$

So for a graph with c components, $n - m + f = c + 1$

Question 7: A graph G is called *outerplanar* if there is an embedding of G in the plane such that every vertex of G lies on the boundary of the outer face.

(a): Prove that G is outerplanar if and only if $G + K_1$ is planar.

Clearly if G is outerplanar, $G + K_1$ is planar (by simply adding in a vertex on the external plane and connecting with arcs to the cycle). So suppose that $G + K_1$ is planar and consider a plane embedding. If $G + K_1$ has 1 face, clearly G is outerplanar. So let C be the cycle on the boundary of the outer face. Let u be the vertex in K_1 . Then either u is on C , or it lies inside the cycle. In both cases, since u is adjacent to every vertex on C , it divides C into triangles. Any other vertex in G must lie in one of those triangles. Any vertex adjacent to both corners of the triangle can join the cycle. Any adjacent to only one, can be placed on the exterior of the cycle by that vertex. Any adjacent to none are disconnected without u and can be placed anywhere on the exterior. Removing u gives us this result. We make the same argument for u on the exterior. This shows G is outerplanar.

(b): If G is an outerplanar graph of order $n \geq 2$ and size m , prove that $m \leq 2n - 3$

By (a), if G is outerplanar, $H = G + K_1$ is planar. Then $n_H \geq 3$, so $m_H \leq 3n_H - 6$. But $m_H = m_G + n_G$ (adding an edge for every vertex in G), so $m_H \leq 3n_H - 6 \iff m_G + n_G \leq 3(n_G + 1) - 6 \iff m_G \leq 2n_G - 3$

(c): Prove that a graph is outerplanar if and only if it contains no subdivision of K_4 or $K_{2,3}$

Suppose that G contains a subdivision H of K_4 . Then $H + K_1 \subseteq G + K_1$ is a subdivision of K_5 . That is, $G + K_1$ is not planar (Kuratowski's Theorem) and thus not outerplanar. Similarly, if G contains a subdivision of $K_{2,3}$, then adding a vertex v , and connecting it to the bipartite set with 3 vertices, this new graph is a subgraph of $G + K_1$ with this new vertex being the K_1 vertex (and only taking some of the new edges). Thus $G + K_1$ is not planar and G is not outerplanar. Now suppose that G is not outerplanar. Then $G + K_1$ is not planar. So $G + K_1$ contains a subdivision of either K_5 or $K_{3,3}$. If the vertex from K_1 is in the subdivision, simply removing it if so, or just removing any vertex (and the connecting edges), we get a subdivision of either K_4 or $K_{2,3}$.

Question 8: For each $k \in \{1, 2, 3\}$, find a nonplanar graph with chromatic number k , or prove that no such graph exists.

For $k = 1$, a graph has chromatic number 1 if it is a collection of isolated vertices. This is clearly planar. For $k = 2$, use graph $K_{3,3}$ which is non-planar and bipartite, so each partite set is a colour. For $k = 3$, start with graph $K_{3,3}$, and add one new vertex connecting to one vertex from each partite set. Call this vertex v . Colouring v with colour 1 forces the two partite sets to have different colours, both not 1. So 3 colours are needed as required.

Question 9: Recall that the average distance of a connected graph G is $\mu(G) = \frac{2}{n(n-1)} \sum_{u,v \in V(G)} d(u, v)$. If G is a maximal planar graph of order n and diameter 2, find a formula for $\mu(G)$

Suppose G is of size m . Since $\text{diam}(G) = 2$, we have that $d(u, v) = 1$ or $d(u, v) = 2$ for $u \neq v$. We have that $u = v$ if and only if $d(u, v) = 0$, and $d(u, v) = 1$ if and only if graph G has edge $uv \in E(G)$. All other pairs of distinct vertices then have distance 2, with the total number of pairs of vertices being $\binom{n}{2} = \frac{n(n-1)}{2}$. So

$$\sum_{u,v \in V(G)} d(u, v) = 1 \cdot m + 2 \cdot \left(\frac{n(n-1)}{2} - m \right)$$

Since G is maximal planar, we have that $m = 3n - 6$ and so

$$\mu(G) = \frac{2}{n(n-1)} \sum_{u,v \in V(G)} d(u, v) = \frac{2}{n(n-1)} [m + n(n-1) - 2m] = \frac{2n(n-1) - 2m}{n(n-1)} = \frac{2n(n-1) - 2(3n-6)}{n(n-1)}$$

which simplifies to

$$\mu(G) = \frac{2n^2 - 8n + 12}{n(n-1)}$$

Question 10: Not Applicable

Question 11: If G is a 2-connected plane graph, prove that the boundary of every face is a cycle.

Assume that we have a face whose boundary is not a cycle. If G contains no cycles, it is a tree and thus has bridges, a contradiction. So the boundary of this face contains a cycle, and an edge not on the cycle. Call this edge e . Then e is a bridge (if $e = uv$, there is no other way for u to reach v other than edge e , otherwise we have another cycle and the boundary of our original face is a cycle), which contradicts that G is 2-connected.

Question 12: Suppose that G is a plane graph in which there are two regions with the same boundary. Prove that G is a cycle.

Let F_1 and F_2 be faces with the same boundary. Every edge lies in this common boundary lies between F_1 and F_2 . Every edge can only separate exactly 2 faces. If we had more than 2 faces, at least 3, there would be an interior edge on the boundary of 2 faces, so they cannot have the same boundary as the external face. By considering the boundary of the external face, we can find an edge on the boundary of one face that isn't on the other. So we have exactly 2 faces, and let the boundary embedding be its largest cycle (possibly with other edges). If there are other edges on the cycle, they are either internal or external (in the cycle or outside). Internal implies that the internal faces boundary includes the edge but not the external face. Similarly, external gives a contradiction. Thus the boundary must just be a cycle.

Question 13: Prove Lemma 4.2.1

Let G be a maximal plane graph of order at least 3 and F a face of G . Consider $G[F]$. The boundary contains some cycle C_k . Let C_k be the cycle contained in the boundary for the smallest possible k . If $k \geq 4$, then we can find vertices x and y on the cycle C_k with $xy \notin E(G)$. Adding edge xy will not break planarity, as if it did, it must intersect another edge internal to our cycle and this would contradict that C_k is the minimal cycle. But since we can add this edge, it contradicts that G is maximal planar. So $G[F] \cong C_3$. From this, if $xy, xz \in E(G[F])$, then since $G[F] \cong C_3$, we must then have $yz \in E(G[F])$

Question 14: Prove that the average degree of the vertices of a planar graph is less than 6

The average degree of a graph G is given by:

$$\frac{\sum_{v \in V(G)} \deg v}{n}$$

By the Handshaking Lemma, $\sum_{v \in V} \deg v = 2m$. If $n \geq 3$, then using that $m \leq 3n - 6$ for planar graphs, we get

$$\frac{\sum_{v \in V(G)} \deg v}{n} = \frac{2m}{n} \leq \frac{6n - 12}{n} < \frac{6n}{n} = 6$$

On the other hand, if $n < 3$, then $n = 1$ or $n = 2$. Both cases are trivial, as the degree of each vertex cannot exceed 1

Question 15: Prove that every 5-connected planar graph has at least 12 vertices.

In a planar graph, $\delta(G) \leq 5$. But $\kappa(G) \geq 5$, by Whitney's Theorem, implies $\delta(G) = 5$. Using the Handshaking Lemma,

$$6n - 12 \geq 2m = \sum_{v \in V} \deg v \geq 5n \Rightarrow n \geq 12$$

Question 16: If G is a planar graph and S a triangle in G (i.e., a subgraph isomorphic to K_3) that is a vertex-cut, then S is called a *separating triangle*. Prove that a minimum counterexample to the Four Colour Theorem cannot contain a separating triangle.

(This question works for the minimum with respect to n, m or f , we will use f). Let G be a counterexample to the Four Colour Theorem with minimal faces. Suppose that G has a separating triangle S . Then $G - S$ has $\leq f - 1$ faces (as it splits into at least 2 components, removing a face and possibly merging of faces). So we can colour $G - S$ with 4 colours. But, since $S \cong K_3$, there are at most 3 faces adjacent to S in G . This means we can colour G with 4 colours (by simply using the colour not used in $G - S$), a contradiction. Thus any minimum counterexample to the Four Colour Theorem cannot contain a separating triangle.

Doing it in terms of vertex-colourings, let T be a separating triangle. Each component in $G - T$ can be coloured with 4 colours. Let $T = \{v_1, v_2, v_3\}$. Since we assume G is minimal, $G - \{v_1\}$ is 4-colourable. So adding v_2, v_3 is 4-colourable. But the component that v_1 connects to is 4-colourable, and $G - v_2$ is also 4-colourable. So we can simply (possibly relabelling colours) colour v_1 in a new way

Question 17: Prove that Every planar graph is 4-colourable and Every maximal planar graph is 4-colourable are equivalent statements (this implies that we may assume that a minimum counterexample to the Four Colour Theorem is a maximal planar graph).

(Note that this is vertex colouring, as a face colouring can be seen as a vertex colouring of a different graph, where the vertices are the faces and the edges are between faces that share a boundary)

Since every maximal planar graph is itself a planar graph, we get that 'Every planar graph is 4-colourable' implies 'Every maximal planar graph is 4-colourable'. On the other hand, if every maximal planar graph is 4-colourable, then every planar graph can be extended to a maximal planar graph and will thus be 4-colourable as a subgraph of a 4-colourable graph.

Question 18: If G is a maximal planar graph with $\delta(G) = 3$, prove that G contains K_4 as a subgraph

Since G is maximal planar, we get $3 \leq \kappa(G) \leq \delta(G) = 3$. So $\kappa(G) = 3$. Let vertex v be of minimal degree, so $\deg v = 3$. Then $|N(v)| = 3$, and so $N(v)$ is a minimal separating set. Then $\delta(G[N(v)]) \geq 2$ and it must follow that $\delta(G[N(v)]) = 2$, so $G[N(v)] \cong K_3$. But then the subgraph $G[N(v)] \cong K_4$, as every vertex has degree 3.

Or use the cycle $C(v)$ such that $V(C) = N(v)$

Question 19: Determine for which values of t the hypercube Q_t is planar.

It is easy to show Q_t is planar for $t = 1, 2, 3$. We will show that Q_4 is not planar, and thus Q_t is not for $t \geq 4$ as it contains Q_4 as a subgraph. We can characterise Q_4 as the graph of all binary strings of length 4, and edges between vertices are those binary strings whose bits differ at exactly one position. We will find a subgraph of Q_4 that is a subdivision of $K_{3,3}$. Take vertex $(0, 0, 0, 0)$ and edges to vertices $(0, 0, 0, 1)$, $(0, 0, 1, 0)$, $(0, 1, 0, 0)$. Then take vertex $(1, 0, 0, 0)$ and paths $(1, 0, 0, 0) \rightarrow (1, 1, 0, 0) \rightarrow (0, 1, 0, 0)$, $(1, 0, 0, 0) \rightarrow (1, 0, 1, 0) \rightarrow (0, 0, 1, 0)$, $(1, 0, 0, 0) \rightarrow (1, 0, 0, 1) \rightarrow (0, 0, 0, 1)$, and finally vertex $(0, 0, 1, 1)$ and paths $(0, 0, 1, 1) \rightarrow (0, 0, 1, 0)$, $(0, 0, 1, 1) \rightarrow (0, 0, 0, 1)$, $(0, 0, 1, 1) \rightarrow (0, 1, 1, 1) \rightarrow (0, 1, 0, 1) \rightarrow (0, 1, 0, 0)$. This subgraph is a subdivision of $K_{3,3}$ as required, so by Kuratowski's Theorem, it is nonplanar.

Question 20: Prove Corollary 4.3.2.

(Rough sketch), Use the Art Gallery Theorem to get a collection of vertices and move them all slightly inwards, by some small $\varepsilon > 0$. The direction of moving can be taken as half the angle of the two vertices adjacent to the original vertex, directed inwards.

Question 21:

N/A